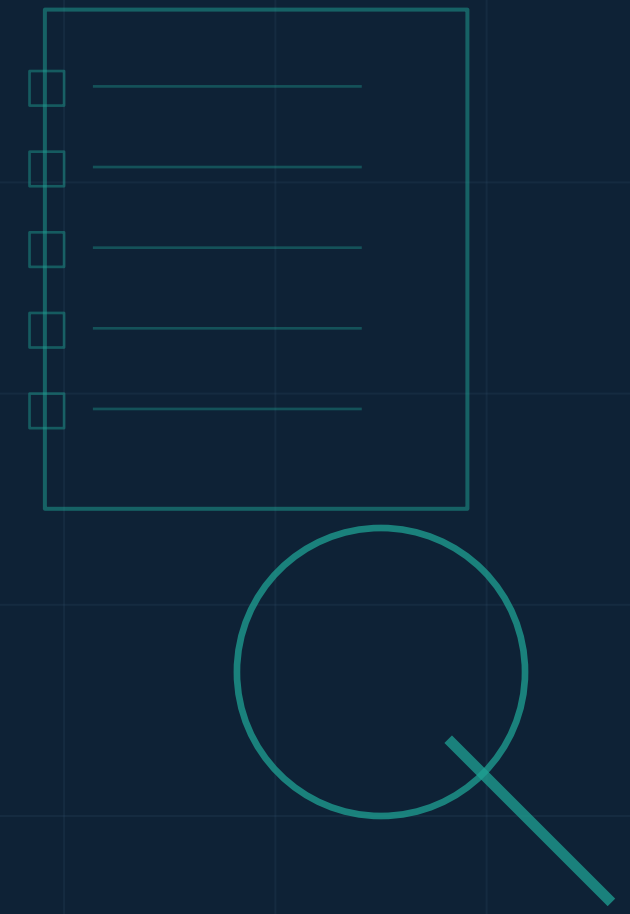
CSE 471 · LECTURE 03

Requirements Determination

what the system must do · and how we find out



Objectives

1

CREATE

Understand how to create a requirements definition.

2

LEARN

Become familiar with requirements analysis techniques.

3

APPLY

Understand when to use each requirements analysis technique.

4

STRATEGIZE

Understand what and where to use requirements analysis strategies.

LECTURE 03

Today's Roadmap

S1

SEGMENT 1

Requirements Specification

What requirements are and the types that matter.

S2

SEGMENT 2

Requirement Gathering & Interviews

Techniques to elicit requirements from people.

S3

SEGMENT 3

Joint Application Design (JAD)

Facilitated workshops to gather requirements together.

S4

SEGMENT 4

Other Gathering Techniques

Questionnaires, document analysis, observation & more.

Key Ideas

The goal

The goal of the analysis phase is to truly understand the requirements of the new system and develop a system that addresses them.

1

1ST CHALLENGE

Collecting and integrating the information.

2

2ND CHALLENGE

Finding the right people to participate.

From Idea to System Proposal

This phase takes the general ideas in the system request and refines them into:

Refines into

1. **Detailed requirements definition (this chapter)**
2. Functional models
3. Structural models
4. Behavioral models

SYSTEM PROPOSAL

This becomes the system proposal, which includes revised project-management deliverables:

1. **Feasibility analysis**
2. **Workplan**

What Is a Requirement?

A requirement is a statement of what the system must do, or what characteristics it needs to have.

BIZ

Business

What the business needs

USER

User

What the users need to do

FUNC

Functional

What the software should do

NON-F

Nonfunctional

Characteristics the system should have

SYS

System

How the system should be built

Functional Requirements

1.0

A process to perform

A process the system should perform as a part of supporting a user task.

2.0

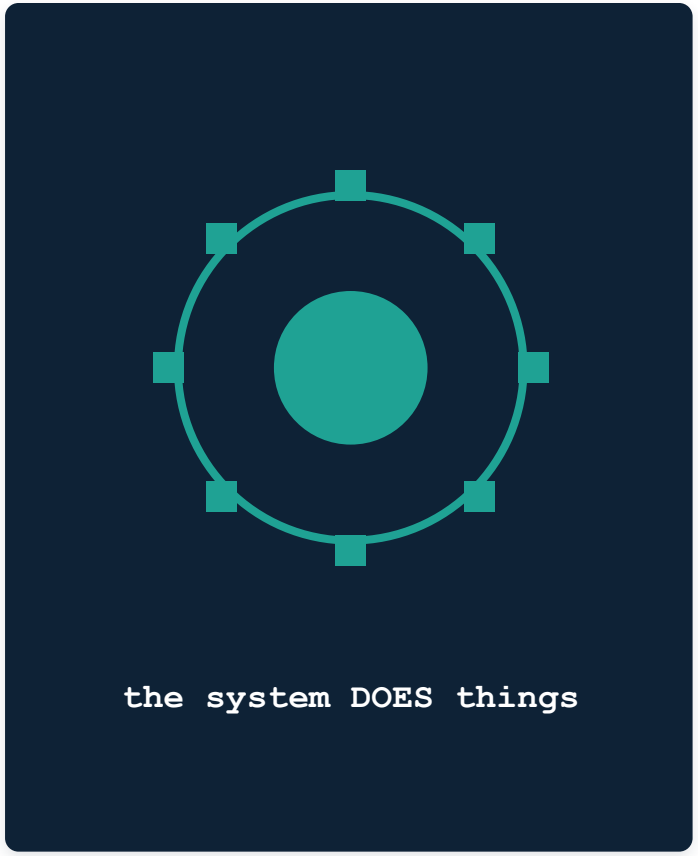
Information to provide

Information the system should provide as the user performs a task.

3.0

Supports the user's work

Specify the support the system will provide to the user in fulfilling his/her work tasks.



Non-Functional Requirements

Nonfunctional requirements are “the quality attributes, design and implementation constraints, and external interfaces which a product must have.”

Behavioral properties the system must have:

OP

Operational

Physical and technical operating environment

PERF

Performance

Speed, capacity, and reliability needs

SEC

Security

Access restrictions, needed safeguards

C&P

Cultural & Political

Issues that will affect the final system

WORKED EXAMPLE

Functional Requirements — Vehicle Dealer

1. New Vehicle Management

- 1.1 View the current new-vehicle inventory.
- 1.2 Allow the manager to place orders for new vehicles.
- 1.3 Record new vehicles added to inventory when received.

2. Vehicle Sales Management

- 2.1 Enable salespersons to create a customer offer.
- 2.2 Show whether an offer is pending on a specific vehicle.
- 2.3 Enable managers to record approval of an offer.
- 2.4 Prepare a sales contract & shop work order.
- 2.5 Record customer deposit, payment & vehicle purchase.

3. Used Vehicle Management

- 3.1 Record information on a customer trade-in vehicle … etc.

WORKED EXAMPLE

Nonfunctional Requirements — Vehicle Dealer

OP Operational

Run on tablet PCs used by salespeople.

Interface with the shop management system.

Connect to printers wirelessly.

PERF Performance

Support a sales staff of 15 salespeople.

Update pending offers on vehicles every 15 minutes.

SEC Security

No salesperson can access another's customer contacts.

Only owner & sales manager may approve offers.

Each tablet restricted to its assigned salesperson.

C&P Cultural & Political

Company policy: all equipment purchased from Dell.

Customer info protected per the Data Protection Act.

System will conform to the state's “lemon law.”

Types of Nonfunctional Requirements

REQUIREMENT	DESCRIPTION	EXAMPLES
Operational	The physical and technical environments in which the system will operate.	Runs on Android mobile devices · Integrates with the existing inventory system · Compatible with any web browser.
Performance	The speed, capacity, and reliability of the system.	Interaction ≤ 2 seconds · New status within 5 min of a change · Available 24/365 · 300 simultaneous users 9–11 a.m., 150 otherwise.
Security	Who has authorized access to the system, and under what circumstances.	Only direct managers see staff records · Technicians see only their own assignments · Safeguards against viruses, worms, Trojans.
Cultural & Political	Cultural, political, and legal factors that affect the system.	Distinguishes US vs. other currencies · Buy computers only from Dell · Country managers authorize custom UIs · Complies with the Data Protection Act.

Functional vs. Non-Functional

#	FUNCTIONAL REQUIREMENT	NON-FUNCTIONAL REQUIREMENT
1	Defines all services/functions the customer requires the system to provide.	Defines system properties & constraints, e.g. reliability, response time, storage, and I/O device capability.
2	Describes what the software should do.	Describes how the software will do it — not what.
3	Related to business (e.g. order-value or payroll calculations).	Related to improving business performance (e.g. security level checks).
4	Functional requirements are easy to test.	Nonfunctional requirements are difficult to test.
5	Related to individual system features.	Related to the system as a whole.
6	Failure may degrade the system.	Failure may make the whole system unusable.

2

SEGMENT 2

Requirements Analysis Strategies

How analysts uncover what the system really needs



Five Elicitation Techniques

Analysts search for requirements using a variety of techniques — making sure current business processes and the needs for the new system are well understood before moving into design.



1

Interviews



2

JAD Sessions



3

Questionnaires



4

Document Analysis



5

Observation

SEGMENT 2

Requirements Analysis Strategies

Seven strategies for turning raw information into real improvements:

1. **Problem Analysis**
2. **Root Cause Analysis**
3. **Duration Analysis**
4. **Activity-Based Costing**
5. **Benchmarking**
6. **Technology Analysis**
7. **Activity Elimination**

STRATEGY 1

Problem Analysis

1.0

Ask the users

Ask users to identify problems and solutions.

2.0

Small & incremental

Improvements tend to be small and incremental.

3.0

Limited payoff

Rarely finds improvements with significant business value.



Simplest strategy — but rarely surfaces big wins.

STRATEGY 2

Root Cause Analysis

1.0

Ask for problems, not solutions

Users give a prioritized list of problems and all possible root causes for them.

2.0

Investigate each cause

Analysts investigate each root cause to find solutions for the highest-priority problems.

3.0

Find shared causes

Look for root causes that are common to multiple problems.



Dig beneath symptoms to the true source.

STRATEGY 3

Duration Analysis

1.0

Time each step

Calculate the time needed for each process step.

2.0

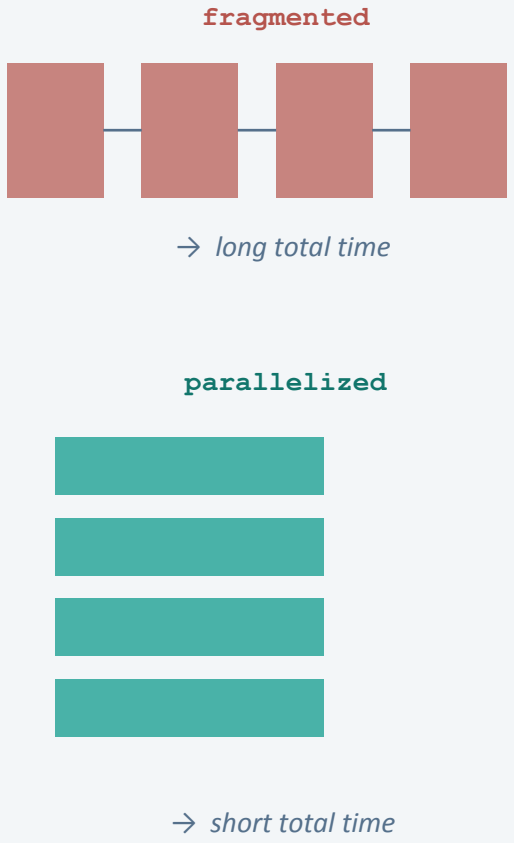
Time the whole

Calculate the time needed for the overall process.

3.0

Compare them

A large difference indicates a badly fragmented process.



Potential solutions

Process integration — Use fewer people, each with broader responsibilities.

Parallelization — Perform individual steps simultaneously.

STRATEGY 4

Activity-Based Costing

1.0

Cost each step

Calculate the cost of each process step.

2.0

Direct + indirect

Consider both direct and indirect costs.

3.0

Target the costly

Identify the most costly steps and focus improvement efforts on them.



Follow the money to the biggest cost drivers.

STRATEGY 5

Benchmarking

1.0

Study others

Study how other organizations perform the same business process.

2.0

Informal benchmarking

Common for customer-facing processes.

3.0

Be the customer

Interact with other businesses' processes as if you are a customer.



Learn from how others already do it.

STRATEGY 6

Technology Analysis

1.0

Analysts list tech

Analysts list important and interesting technologies.

2.0

Managers list tech

Managers list important and interesting technologies.

3.0

Map to the business

The group identifies how each might be applied and how the business might benefit.



Let new technology suggest new possibilities.

STRATEGY 7

Activity Elimination

1.0

Imagine removing it

Identify what would happen if each organizational activity were eliminated.

2.0

Force-fit the test

Use “force-fit” to test all possibilities — even ones that seem essential.



If we cut this, what truly breaks?

Comparing Analysis Techniques

When choosing a strategy, weigh it across four dimensions:

VALUE

Potential Business Value

How much improvement could it unlock?

COST

Project Cost

How expensive is the technique to run?

BREADTH

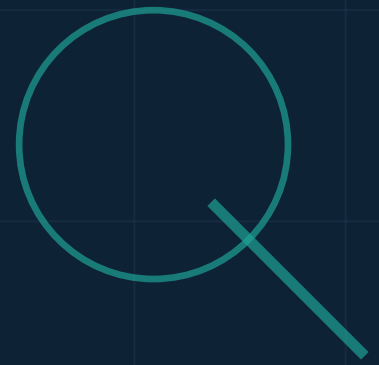
Breadth of Analysis

How wide a scope does it cover?

RISK

Risk

How likely is it to mislead or fail?



THANK YOU

Requirements Determination

Specify what the system must do · gather it with the right technique · pick the strategy that fits the project.

S1	Requirements Specification
S2	Gathering & Interviews
S3	JAD
S4	Other Techniques